

2.5 Truth Tables

Truth tables are used in logic to show the truth values of statements. They help us understand how logical expressions work.

- A truth table lists all possible combinations of truth values.
- It shows the final result for each combination.
- Truth tables are important in propositional logic, computer science, and mathematics.
- They help in checking the correctness of logical expressions.
- Truth tables make logical thinking clear and simple.

Example Truth Table:

A	B	A AND B	A OR B	NOT A
True	True	True	True	False
True	False	False	True	False
False	True	False	True	True
False	False	False	False	True

Figure 2.5: Truth Table for Logical Expressions

Purpose of Truth Tables

- Show how a logical expression behaves
- Help find when a statement is true or false
- List all possible input values
- Remove confusion in logical reasoning
- Allow checking of complex expressions easily
- Help compare different logical statements
- Useful in decision-making

Creating Truth Tables for Logical Expressions

1. Identify the propositions.
2. Give each proposition a truth value of True or False.
3. List all possible combinations of truth values.
4. Apply logical connectives step by step.
5. Write the final result for each row.

Truth tables must be neat and organized.

Truth Values of Propositions

Each proposition P or Q can take the value **True (T)** or **False (F)**. Truth tables enumerate all possible combinations (2^n rows for n propositions). The truth value of a compound proposition is determined by the values of its components.

Short Questions

1. What is a truth table?
2. What is the main purpose of a truth table?
3. How many rows will a truth table have for 2 propositions?
4. How many rows will a truth table have for 3 propositions?
5. Write the steps to create a truth table.
6. What are the possible truth values of a proposition?
7. Why are truth tables important in computer science?
8. Draw the truth table for AND operation.
9. Draw the truth table for OR operation.
10. Draw the truth table for NOT operation.